Finite terms: DGLAP, running coupling and the finite terms

1 What is done here

- Starting point: the remaining terms of `Finite_terms_split.pdf` (Secs. 7–9), i.e. every row of `Finite_terms_1.tex` without its BFKL ($\log \frac{\vee}{k^+}$) terms, with $\left[\frac{1}{p^+}\right]_+$ kept as $\left[\frac{1}{p^+}\right]_+^{(k^+)}$.
- Only three of them contain $\frac{1}{\epsilon}$ poles or β_0: Group I Row II part 1, Group I Row V 2b UV, Group II Row IV. All other rows are finite.
- From these three rows, and the coupling counterterm of LO, the running coupling (Sec. 2) and the DGLAP term (Sec. 3) are taken out. What is left, with all other rows, are the finite terms (Sec. 4). Sec. 8 is the result, Sec. 9 the comparison with the paper.

Notation. $a \equiv \frac{k^+}{\vee}$, $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $d_\perp = 2-\epsilon$, $c_0 \equiv 2e^{-\gamma_E}$, $\ell_r \equiv \log\frac{r^2\mu^2}{c_0^2} = 2\gamma_E + \log\frac{r^2\mu^2}{4}$, $\hat{E}_r \equiv \frac{2}{\epsilon} + \ell_r$; $X = x-y$, $X' = x'-y'$, $\Delta = y'-y$. LO in your notation, and LO with a weight f:

$$\mathrm{LO}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f\, \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x),$$

$\mathrm{LO} \equiv \mathrm{LO}[1]$, and $\mathrm{LO}_\xi[f]$ = the same with $e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$. $P^{(r)}_{gg,+}(\xi) = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big]$ with the plus prescription on $[a,1]$, $P_{gg} = P^{(r)}_{gg,+} + \frac{\beta_0}{2}\delta(1-\xi)$. The conversion of the three rows to units of LO was checked against your file before (`../finite_v2/final_checks.json`); the bookkeeping below is checked in `checks.py`.

2 Running coupling

The β_0 terms. Two rows have β_0: the $\frac{11}{3}$ of Group I Row II part 1 and the $-\frac{11}{6}$ of Group I Row V 2b UV:

$$\frac{d^3N_1^{\beta_0}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\frac{11}{3}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon} - 2\gamma - \log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\big] \quad + \text{c.c.} \;=\; \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\hat{E}_X + \hat{E}_{X'}\big], \tag{2.1}$$

$$\frac{d^3N_{2b\,\mathrm{UV}}^{\beta_0}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\frac{11}{6}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y) \quad + \text{c.c.} \;=\; -\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\hat{E}_X + \hat{E}_{X'}\big]. \tag{2.2}$$

With $y \to x$, $z \to w$ in (2.2), the measure, the phase, the kernel and the bracket are those of (2.1), the colour factor changes sign, and $\frac{g^4N_c}{16\pi^4}\frac{11}{6} = \frac{g^4N_c}{32\pi^4}\frac{11}{3}$. So

$$\boxed{\frac{d^3N_1^{\beta_0}}{d^3k} + \frac{d^3N_{2b\,\mathrm{UV}}^{\beta_0}}{d^3k} = 0} \tag{2.3}$$

for every a: pole, γ_E, μ and $\log(x-w)^2$ all cancel (`checks.json`). The virtual correction (Row II) and the real emission at short distance (Row V 2b UV) carry opposite β_0 terms, and no $\beta_0\log(x-w)^2$, i.e. no running of the coupling at the emission distance, is left.

The coupling of LO. LO is $\propto g^2$. In $\overline{MS}$, $g^2 \to g^2(\mu_R)\big[1 - \frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\big]$ ($d_\perp = 2-\epsilon$), which adds

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{CT}} = -\frac{\alpha_s\beta_0}{4\pi}\,\frac{2}{\epsilon}\,\mathrm{LO} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{\beta_0}{2}\,\mathrm{LO}. \tag{2.4}$$

This pole is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of the DGLAP pole (Sec. 3). After the coupling is renormalized at μ_R and the fragmentation function at μ_F (both $\overline{MS}$), the only β_0 term of the result is the renormalization-group term

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\beta_0} = \frac{\alpha_s\beta_0}{4\pi}\,\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO}\Big|_{\alpha_s(\mu_R)}} \tag{2.5}$$

It makes the result independent of μ_R: $\frac{d}{d\log\mu_R^2}\mathrm{LO}|_{\alpha_s(\mu_R)} = -\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}$. It vanishes for $\mu_R = \mu_F$; then LO is taken with $\alpha_s(\mu_F)$.

The constant. The non-logarithmic part of the bracket of Row II part 1, $-\frac{67}{9}+\frac{\pi^2}{3}$, gives

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\Big[-\frac{67}{9}+\frac{\pi^2}{3}\Big]$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big]+\text{c.c.} \;=\; \boxed{\frac{\alpha_s}{2\pi}\,N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\text{LO}} \tag{2.6}$$

$N_c(\frac{67}{9}-\frac{\pi^2}{3})=2K$ with $K=N_c(\frac{67}{18}-\frac{\pi^2}{6})$, the constant of the two-loop cusp anomalous dimension that the CMW scheme absorbs into the coupling. It is the finite companion of β_0 in this row.

3 DGLAP

All poles. The poles of the remaining terms, in units of LO (`checks.json`):

G1 Row II part 1 + c.c.	$\frac{\alpha_s N_c}{4\pi}\frac{4}{\epsilon}\big[\frac{11}{3}+\log(1-a)\big]\text{LO}$
G1 Row V 2b UV + c.c.	$\frac{\alpha_s N_c}{2\pi}\frac{4}{\epsilon}\big[-\frac{11}{6}+2\log(1-\frac{\Lambda}{k^+})\big]\text{LO}\quad(\log(1-\frac{\Lambda}{k^+})\to 0)$
G2 Row IV, lines 1, 2	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi$
coupling counterterm (2.4)	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\frac{\beta_0}{2}\text{LO}$
sum	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\Big[\int_a^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\text{LO}_\xi-N_c\log(1-a)\,\text{LO}\Big]$

The first term is the DGLAP pole: it is absorbed in $D_{h/g}(\zeta,\mu_F)$ ($\overline{MS}$), with the convolution $\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta,\mu_F)\frac{d^3N_g}{d^3k}(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$. The second is not (end of this section).

The DGLAP term. Group II Row IV, lines 1 and 2 of the remaining terms, with the pole removed (your bracket $\big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\big]=-\hat{E}_\Delta+O(\epsilon)$ becomes $-\ell_\Delta$; the -1 cancels against $-\frac{d_\perp}{\epsilon}$), together with the $\log\frac{\vee}{k^+}H_4$ subtraction:

$$\frac{d^3N_{\text{DGLAP}}}{d^3k}=\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-2\gamma_E-\log\left(\frac{(y'-y)^2\,\mu_F^2}{4}\right)\right]$$
$$\times\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\xi(1-\xi)+\frac{1-\xi}{\xi}\right]+\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}\right.$$
$$\left.+\log\left(1-\frac{k^+}{\vee}\right)e^{-ik\cdot(y'-y)}\right]$$
$$\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\,[U^{ec}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)$$
$$-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}$$
$$\times\Big[U^{e'c}(z)-U^{e'c}(x')\Big]\,[U^{ec}(z)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)$$
$$=\boxed{-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\big[\ell_\Delta(\mu_F)\big]-\log\frac{\vee}{k^+}\,H_4,}\qquad \ell_\Delta(\mu_F)=\log\frac{(y'-y)^2\mu_F^2}{c_0^2}. \tag{3.1}$$

- The bracket in ξ is the plus distribution $\frac{1}{2N_c}P^{(r)}_{gg,+}$ on $[\frac{k^+}{\vee},1]$; its last term is your $\log(1-\frac{k^+}{\vee})$ line.
- $\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g}=\frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}$ cancels the μ_F dependence of $\ell_\Delta(\mu_F)$ and of (2.5).
- The last two lines ($-\log\frac{\vee}{k^+}H_4$) remove the $\log\frac{\vee}{k^+}$ that the ξ integral produces at $\xi\to\frac{k^+}{\vee}$ (it is in the BFKL terms); with them the DGLAP term is finite for $\vee\to\infty$.
- Line 3 of the row (linear polarization) is finite and not DGLAP; it is in Sec. 4.

The pole that is left (as before, Q1). The second term of the pole, with its finite partner from Row II part 1:

$$\frac{d^3N}{d^3k}\bigg|_{\text{Q1}}=\frac{\alpha_sN_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\,\text{LO}\big[\hat{E}_X+\hat{E}_{X'}\big]. \tag{3.2}$$

It is not DGLAP (it is not $P_{gg}\otimes$LO) and not running coupling. It comes from the limit $\xi\geq\frac{k^+}{\vee}$ in the virtual term of Row II part 1 and vanishes for $k^+\ll\vee$, i.e. at mid rapidity in the high-energy limit. In AABKL the virtual term has $\int_0^1 d\xi$ and there is no such term (Sec. 9).

4 The finite terms

The remaining terms of `Finite_terms_split.pdf`, Secs. 7–9, without the running coupling and the DGLAP term. Group I Row II part 1, Row V 2b UV and Group II Row IV are reduced as described under each; every other row is as in `Finite_terms_split.pdf` (blue: changed with respect to `Finite_terms_1.tex`). All of them are finite for $\epsilon \to 0$, $\vee \to \infty$ and $\Lambda \to 0$; the terms with $\log(1 \pm \frac{\Lambda}{k^+})$ vanish for $\Lambda \to 0$.

5 Finite terms: Group I (with c.c.)

5.1 Row I

NA

5.2 Row II

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\,\log\left[\frac{\vee}{k^+}\right]\right]$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\right] \tag{5.1}$$

Part 1: what is left is $V_\vee = -\frac{\alpha_s}{2\pi}N_c\log(1-\frac{k^+}{\vee})\log(\frac{\vee}{k^+}-1)$ LO (with c.c.); it vanishes for $k^+ \ll \vee$. The $\frac{11}{3}$ is in Sec. 2 (cancels), the constant $-\frac{67}{9}+\frac{\pi^2}{3}$ in Eq. (2.6), the pole bracket times $\log(1-\frac{k^+}{\vee})$ in Eq. (3.2).

d^3N_{2a}: no remaining term (it is proportional to $\log\frac{\vee}{k^+}$; all of it is in the BFKL terms, `Finite_terms_split.pdf`, Sec. 2).

$$\frac{d^3N_{2\,b}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right] \tag{5.2}$$

where

$$R^i = -2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}\,p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right]-\frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right). \tag{5.3}$$

Part 2b: no change.

5.3 Row III

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]$$
$$+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2\right]}{\left[\left(1+\frac{\Lambda}{k^+}\right)(x-z)^2-(x-w)^2\right]}\right]$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]$$
$$-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1-\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]$$
$$+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left[\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right]$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]$$
$$-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\left[\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2\right]}{\left[(x-w)^2-\frac{\Lambda}{k^+}\,(x-z)^2\right]}\right]$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \tag{5.4}$$

$$\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big]\Bigg\}
\end{aligned} \tag{5.5}$$

$$\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&-\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big] \\
&+\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\big]
\end{aligned}$$

$$
- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+ - (x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+ - (x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+ - (x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+ - (x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+ - (x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+ - (x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+ - (x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+ - (x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-z)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+ - (x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+ - (x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-w)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+ - (x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+ - (x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{5.6}
$$

$$
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{
$$
$$
- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
- \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\left[k^+\,(x-w) - (k^+-\Lambda)\,(x-z)\right]^2}{\left[k^+\,(x-w) - \Lambda\,(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
+ \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\left[k^+\,(x-w) - (k^+-\Lambda)\,(x-z)\right]^2}{\left[k^+\,(x-w) - \Lambda\,(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]
$$
$$
- \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\left[k^+\,(x-w) - (k^+-\Lambda)\,(x-z)\right]^2}{\left[k^+\,(x-w) - \Lambda\,(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2}
$$
$$
\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{5.7}
$$

where

$$\mathcal{A}=\frac{1}{\sqrt{(x-w)^2\,(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\left\{\arctan\left[\frac{(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2\,(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\right]\right.$$
$$\left.-\arctan\left[\frac{\Lambda\,(x-z)^2-k^+\,(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2\,(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\right]\right\}$$

5.4 Row IV

$$\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\left\{\right.$$
$$\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)$$
$$\left.+\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[1-\frac{k^+}{\vee}\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\right\}\times$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\right]\qquad(5.8)$$

5.5 Row V

First part

$$\frac{d^3N_{1\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times$$
$$\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\left[\frac{1}{q^+}\right]_+$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\qquad(5.9)$$

Second part

$$\frac{d^3N_{2a\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.$$
$$\left.+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\right.$$

$$
\begin{aligned}
&- \left[\frac{k^+}{p^+}\right]_+ \left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2} + \frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right] + \left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2} + \frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right] \\
&+ \frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2\,(y-z)^2} - \frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right] \\
&- \left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2} - \frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2} + \frac{(y-z)^m\,(x-z)^i}{(x-z)^2} + \frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\times \\
&\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
&+ \frac{1}{(2\pi)^3}\int_{k^+}^0 \frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{\left[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right] - \left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times \\
&\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right. \\
&\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2} \\
&\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{5.10}
$$

$d^3N_{2b\,\mathrm{UV}}$: nothing is left. Its $-\frac{11}{6}$ is in Sec. 2 (cancels against Row II part 1); its $2\log(1-\frac{\Lambda}{k^+})$ term vanishes for $\Lambda \to 0$.

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+(y-x)^2+(k^+-p^+)(y-z)^2\right]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right. \\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right] + \left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right] \\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right] \\
&\times\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[N_c\,U^{af}(y) - N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{\left[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2\right]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times \\
&\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[N_c\,U^{af}(y) - N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^\Lambda\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{\left[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2\right]} \\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right] \\
&\left[U^{ab'}(x') - U^{ab'}(w')\right]\left[N_c\,U^{af}(y) - N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.- \frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)\\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+ \frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)\\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\log\left[1-\frac{\Lambda}{k^+}\right]\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y)
\end{aligned}
\tag{5.11}
$$

6 Finite terms: Group II

6.1 Row I

6.1.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{\\
&- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&+ \frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\\
&\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^{b}(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\Big]\\
&+ \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{\\
&\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w')^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]\\
&- \frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}\\
&+ \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}\\
&\times\Big[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\Big]\times\Big[U^{ac}(w)\,\{\rho^{c}(x'),\rho^{b}(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^{d}(x'),\rho^{e}(y)\}\Big]
\end{aligned}
\tag{6.1}
$$

6.1.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\left\{\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\right.\\
&\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.\\
&\left.\left.+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}\right.\right.
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}\\
&-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\Bigg]\times\\
&\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{k^+}{\vee}(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{\Lambda\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{6.2}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\\
&\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\\
&\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{6.3}
$$

$$
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{6.4}
$$

$$
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]\times$$

$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$

$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{6.5}$$

$$\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}$$

$$\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$

$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]$$

$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$

$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$

$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$

$$\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$

$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times$$

$$\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w)^2}\right]\times$$

$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$

$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$

$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}$$

$$\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.$$

$$\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times$$

$$\log\left[\frac{\frac{k^+}{\vee}(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}{(x'-w')^2+\frac{\Lambda}{k^+}(x'-z)^2}\right]\times$$

$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$

$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{6.6}$$

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w'\right.}$$

$$\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]$$

$$-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2+(x-w)^2}\right]\times
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\left[\frac{1}{\vee}-\frac{1}{k^++\Lambda}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}
$$

$$
\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}
$$

$$
\times\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]\times
$$

$$
\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{6.7}
$$

6.2 Row II

NA

6.3 Row III

NA

6.4 Row IV

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]
$$

$$
\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{6.8}
$$

Row IV. Only line 3 (linear polarization) is left here; it is finite. Lines 1, 2 and the log $\frac{\vee}{k^+}H_4$ subtraction are the DGLAP term, Eq. (3.1); their pole is in Sec. 3.

7 Finite terms: Group III (with c.c.)

7.1 Row I

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{
$$

$$
-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$

$$
\left.-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]
$$

$$
+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}
$$

$$
\begin{aligned}
&- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{7.1}
$$

7.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'}e^{-ik(w'-w)}\Bigg\{&\\
&-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}\,(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.\\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{7.2}
$$

7.3 Row III

7.3.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
\end{aligned}
\tag{7.3}
$$

The $\left[\frac{1}{p^+}\right]_+$ term is kept unevaluated; only its subtraction is moved (blue, definition in `Finite_terms_split.pdf`, Sec. 6), $\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+F=\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+^{(k^+)}F-F(0)\log\frac{\vee}{k^+}$ with $F(p^+)=e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}$, $F(0)=1$. The $-F(0)\log\frac{\vee}{k^+}$ gives the BFKL term of this row (`Finite_terms_split.pdf`, Sec. 4).

7.3.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2\,\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,[(x-z)^2-(x-w)^2]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.\left.-\frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right]\right.\\
&\left.-\left[\frac{1}{p^+}\right]_+^{(k^+)}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right.
\end{aligned}
$$

$$
-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\times
$$

$$
\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x}e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times
$$

$$
\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{7.4}
$$

The $\left[\frac{1}{p^+}\right]_+$ term is kept unevaluated; only its subtraction is moved (blue, definition in `Finite_terms_split.pdf`, Sec. 6), with $F(p^+)=\frac{e^{-ik\cdot(y'-z)p^+/k^+}}{p^+(x-z)^2+k^+(x-w)^2}$, $F(0)=\frac{1}{k^+(x-w)^2}$. The $-F(0)\log\frac{\vee}{k^+}$ gives the BFKL term of this row (`Finite_terms_split.pdf`, Sec. 4).

7.4 Row IV

NA

7.5 Row V

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times
$$

$$
\left[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[U^{e'c'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\right] \tag{7.5}
$$

The $\left[\frac{1}{p^+}\right]_+$ term is kept unevaluated; only its subtraction is moved (blue, definition in `Finite_terms_split.pdf`, Sec. 6), $\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+F=\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+^{(k^+)}F-F(0)\log\frac{\vee}{k^+}$ with $F(p^+)=e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}$, $F(0)=1$. The $-F(0)\log\frac{\vee}{k^+}$ gives the BFKL term of this row (`Finite_terms_split.pdf`, Sec. 4).

7.6 Row VI

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times
$$

$$
\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x)-U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x)-U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right] \tag{7.6}
$$

The $\left[\frac{1}{p^+}\right]_+$ term is kept unevaluated; only its subtraction is moved (blue, definition in `Finite_terms_split.pdf`, Sec. 6), $\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+F=\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+^{(k^+)}F-F(0)\log\frac{\vee}{k^+}$ with $F(p^+)=e^{-ik\cdot(y'-z)\frac{p^+}{k^+}}$, $F(0)=1$. The $-F(0)\log\frac{\vee}{k^+}$ gives the BFKL term of this row (`Finite_terms_split.pdf`, Sec. 4).

8 Result

$$
\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3}\, D_{h/g}(\zeta,\mu_F)\, \frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO} + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO} \qquad \text{(running coupling, Sec. 2)}\\
&\quad + \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big] \qquad \text{(evolution: JIMWLK, BFKL)}\\
&\quad + \frac{d^3N_{\mathrm{DGLAP}}}{d^3k} \qquad \text{(Eq. (3.1))}\\
&\quad + \frac{\alpha_s N_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big] \qquad \text{(Q1, Eq. (3.2))}\\
&\quad + \text{finite terms (Sec. 4)} + \text{four-}\rho\text{ term (Q2)} + O(\alpha_s^3).
\end{aligned}
\tag{8.1}
$$

- $\mathcal{J}$, $\mathcal{B}$: the brackets of Secs. 3 and 4 of `Evolution_terms.tex`; $\log\frac{\vee}{\Lambda}\mathcal{J}$ (`Evolution_terms.tex`) plus the BFKL terms $\log\frac{\vee}{k^+}(\mathcal{B}-\mathcal{J})$ (`Finite_terms_split.pdf`).
- The finite terms contain three-ρ rows. Their fully symmetric part needs $\langle\rho\rho\rho\rangle$ (zero for a Gaussian projectile); their two-ρ parts are written out in `Finite_terms_simplified.pdf`, Sec. 9.
- The four-ρ finite term is not in `Finite_terms_1.tex` (`Finite_terms_simplified.pdf`, Sec. 10).

9 Comparison with AABKL (arXiv:1411.2869)

T. Altinoluk, N. Armesto, G. Beuf, A. Kovner, M. Lublinsky: forward single-inclusive production in the hybrid formalism. The projectile is a collinear parton with PDF $f(x_p)$, LO is $O(\alpha_s^0)$, NLO is one splitting. The relevant channel is $g\to g$, their Eqs. (4.20), (4.22) and App. C. Dictionary: $\xi_{\mathrm{AABKL}} = 1-\xi$ (they use the fraction of the unobserved gluon), $x_p \leftrightarrow \frac{k^+}{\vee}$, $A^i(y-z) = -\frac{1}{2\pi}\frac{(y-z)^i}{(y-z)^2} \leftrightarrow -\frac{1}{2\pi}\mathcal{K}(y-z)$, $s_A[y,\bar{y}] \leftrightarrow$ the LO colour factor, $(y,\bar{y}) \leftrightarrow (y,y')$.

1. DGLAP term: the same.

- Splitting function: their $\frac{1-\xi}{[\xi]_+} + \frac{\xi}{1-\xi} + \xi(1-\xi)$ at $\xi\to 1-\xi$ is $\frac{1}{2N_c}P^{(r)}_{gg,+}$, with the plus prescription at the soft end in both (`checks.json`: difference 0).
- Kinematic limit: their real term has $\int_0^{1-x_p} d\xi$, yours $\int_{k^+/\vee}^1 d\xi$; both produce $\log(1-x_p)$, resp. $\log(1-\frac{k^+}{\vee})$, through the plus distribution.
- The logarithm: their after-target term $2C_A\big[A^i(y-z)A^i(\bar{y}-z) - (A^i(z)A^i(z))_{\mu^2}\big](1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar{y}]$ integrates over z to $\log\big((y-\bar{y})^2\mu^2\big)$ times the rescaled dipole; yours is $\ell_\Delta = \log\frac{(y'-y)^2\mu_F^2}{c_0^2}$ times LO_ξ (the rescaling is your $\frac{1}{\xi}$ in the phase). Their App. C, Eq. (7.13): $-\frac{g^2}{2(2\pi)^2}\log\big(\mu_{CO}^2(y-\bar{y})^2\big)P\otimes\ldots$, the same structure as Eq. (3.1).
- Scheme: their Eq. (7.22), $R = 2e^{\Psi(1)} \simeq 1.1229$ relates their scheme to $\overline{MS}$. This is exactly your $c_0 = 2e^{-\gamma_E}$ (`checks.json`: $R - c_0 = 0$), so their $\log(r^2\mu^2)$ and your $\ell_r = \log\frac{r^2\mu^2}{c_0^2}$ are the same $\overline{MS}$ logarithm.
- Hadron convolution: $\int_{x_F}^1 \frac{d\zeta}{\zeta^2}D(\zeta)\frac{d\bar{\sigma}}{d^2p}(\frac{p}{\zeta},\frac{x_F}{\zeta})$ vs. $\int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^3}D_{h/g}(\zeta)\frac{d^3N_g}{d^3k}(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$: the same.

2. Running coupling: different, because LO is different.

- AABKL: LO is $O(\alpha_s^0)$, so there is no coupling to renormalize and no $\beta_0\log\mu_R^2$ term. β_0 enters only as the $\frac{\beta_0}{2}\delta(1-\xi)$ part of P_{gg} in the DGLAP evolution of f^g and D^g. It comes from their virtual term: $\int_0^1 d\xi\big[\frac{1-\xi}{[\xi]_+} + \frac{\xi}{[1-\xi]_+} + \xi(1-\xi)\big] = -\frac{11}{6}$, and $2C_A\cdot\frac{11}{6} = \beta_0$ (`checks.json`).
- Yours: LO is $O(\alpha_s)$. The β_0 of the virtual term (Row II) cancels against the short-distance real emission (Row V 2b UV), and the $\frac{\beta_0}{2}\delta(1-\xi)$ of P_{gg} comes from the coupling counterterm of LO. In the end the same P_{gg} is absorbed in $D_{h/g}$.
- The RG term (2.5) and the constant $\frac{\alpha_s}{2\pi}N_c(\frac{67}{9}-\frac{\pi^2}{3})$LO have no counterpart in AABKL.

3. The finite terms: same colour structures and topologies, different kernels.

AABKL, Eq. (4.22)	yours, Sec. 4	comparison
before the target: $2C_A[A\cdot A - (AA)_{\mu^2}]\,s_A[y,\bar{y}]$	G2 Row I part 2, colour $ff\,U(w')U(w)$	same topology and colour; theirs collinear divergent (PDF subtraction), yours finite: the parent is emitted at $\lvert x-w\rvert$ and the energy denominators are $p^+(x-z)^2 + k^+(x-w)^2$

AABKL, Eq. (4.22)	yours, Sec. 4	comparison
interference: $\frac{N_c^3}{N_c^2-1} A\cdot A\{sss+\cdots-\frac{1}{N_c^2}\overline{X}\}$	G3 Row III part 2; mixed colour terms of G2 Row I part 2	same topology (parent at $(1-\xi)y+\xi z$ vs. your y'), colour the same after projection, kernels differ
virtual: $-x_p f\int_0^1 d\xi(\dots)\{C_A[AA+AA-2(AA)_{\mu^2}]s_A+\dots\}$	G1 Row V 2a, 2b finite: colour $-ff\,U(x)U(z)+N_c U(x)$	colour the same; theirs $\int_0^1 d\xi$, yours $\int_0^{k^+} dp^+$; your Row II has $\xi\geq\frac{k^+}{\vee}$ (Q1, $V_\vee$)
last term: $\int\frac{d\xi}{\xi}[A_\xi-A_\xi]^2\{\dots\}$ (Ioffe time)	evolution, $\log\frac{k^+}{\Lambda}\mathcal{J}+\log\frac{\vee}{k^+}\mathcal{B}$	same soft real−virtual (BK/JIMWLK) structure; theirs regulated by the Ioffe time, yours by the p^+ limits
—	linear polarization (G2 Row IV, line 3)	absent: their incoming gluon is unpolarized
—	parent emitted after the target by the rotated charge (G2 Row I part 2, G3 Rows V, VI)	absent: their parent exists before the target
—	G1 Rows III, IV, G2 Row I part 1, G3 Rows I, II (emission by the charges around the splitting)	absent: no colour charges in their projectile
—	three-ρ, four-ρ terms	absent: one parton
—	$\log\frac{\vee}{k^+}\mathcal{B}$ (BFKL)	absent: their projectile evolution sits in the PDF
—	$V_\vee$, Q1	absent: their virtual term has no x_p limit

In short.

- **DGLAP**: the same as AABKL, in the splitting function, the plus prescription, the kinematic limit, the logarithm and the scheme constant ($R=c_0$).
- **Running coupling**: none in AABKL (LO is α_s^0). Yours has the RG term and the constant $\frac{67}{9}-\frac{\pi^2}{3}$; the β_0 of P_{gg} is the same in both.
- **Finite terms**: they resemble AABKL in colour structure and topology (before the target, interference, virtual), not in the kernels. The difference is the projectile: colour charges that emit the gluon at LO, so your kernels contain the emission distance and there is no collinear divergence to subtract into a PDF. Linear polarization, emission after the target, the terms with emission by the charges around the splitting, three- and four-ρ terms, $V_\vee$ and Q1 have no counterpart.